

Exam questions – Web systems and technologies

1. Create the page below using HTML and CSS (10 points):

PROGRAMMING

Programming is an art that requires *creativity* and logic. To succeed in this field, it is important to learn continuously and practice.

In the modern world, programming skills are increasingly *in demand* and this field is constantly evolving.

2. Create the **form** below using HTML and CSS: (10 points)

Name:

Email:

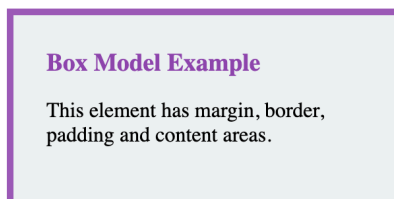
City:

New York ▾

3. Create the **flexbox** layout below using HTML and CSS: (10 points)



4. Use the **box-model** properties to create the following web page fragment: (10 points)



5. Create the page below using HTML and CSS (10 points):

Web Design

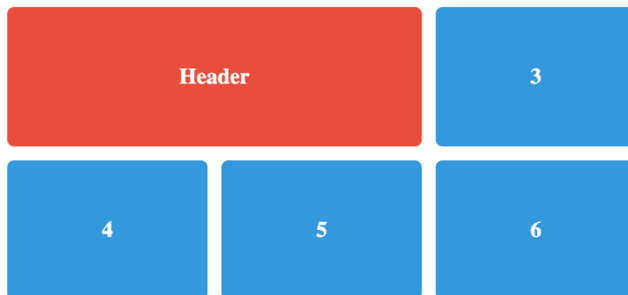
Web design is very important for *user experience*. Good design ensures the success of the site.

The choice of colors, fonts and layout creates a *professional* look.

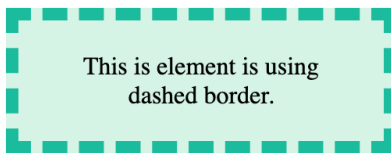
6. Create the **form** below using HTML and CSS: (10 points)

Registration Form

7. Create the **grid** layout below using HTML and CSS: **(10 points)**



8. Use the **box-model** properties to create the following web page fragment: **(10 points)**



9. Create the page below using HTML and CSS. The font-family is 'Courier New'. (10 points):

Frontend Development

Frontend development consists of `HTML, CSS and JavaScript` technologies.

This field requires `creativity` and technical knowledge.

10. Create the **form** below using HTML and CSS: **(10 points)**

Full Name:

Email:

Message:

11. Create the **flexbox** layout below using HTML and CSS: (10 points)

Element 1

Element 2

Element 3

12. Use the **box-model** properties to create the following web page fragment: (10 points)

Card Design

Lorem ipsum dolor, sit amet
consectetur adipisicing elit.

13. Create the page below using HTML and CSS. (10 points):

Web Technologies

Modern web technologies allow creating dynamic and interactive websites.

HTML provides structure, CSS provides design, and JavaScript provides functionality.

14. Create the **form** below using HTML and CSS: (10 points)

Login

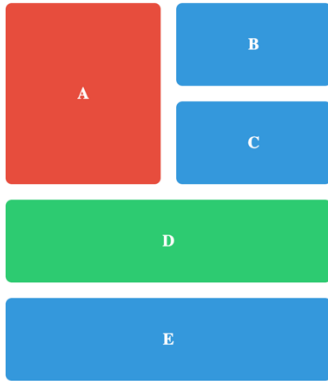
Your email address

Password

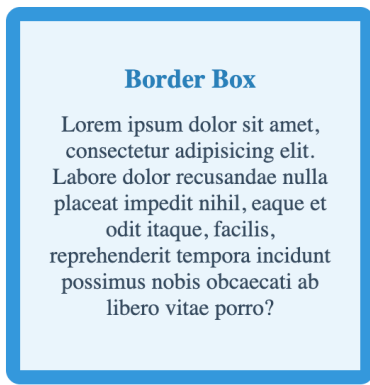
Confirm password

Login

15. Create the **grid** layout below using HTML and CSS: (10 points)



16. Use the **box-model** properties to create the following web page fragment: **(10 points)**



17. Create the page below using HTML and CSS **(10 points)**:

Web Development

Learning web development requires dedication and practice. Understanding **HTML and CSS** is the foundation of creating beautiful websites.

Modern web technologies enable developers to build **responsive and interactive** user interfaces that work across all devices.

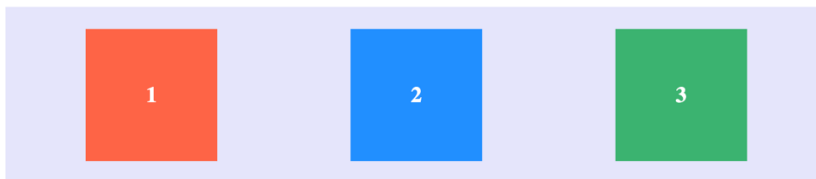
18. Create the following form using HTML and CSS (10 points):

Contact Form
Full Name:

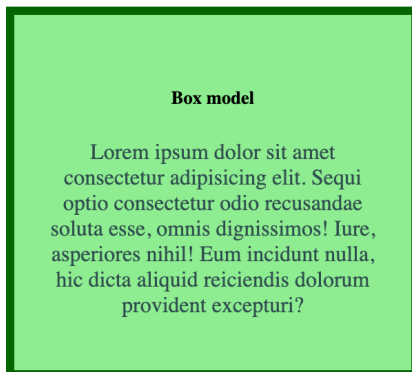
Email Address:

Phone Number:

19. Create the following layout using HTML and CSS flexbox (10 points):



20. Use the **box-model** properties to create the following web page fragment: (10 points)



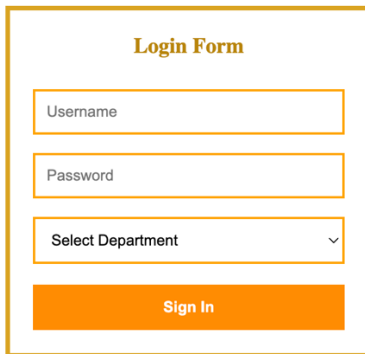
21. Create the page below using HTML and CSS (10 points):

FRONTEND DESIGN

Creating beautiful user interfaces requires an understanding of *design principles* and attention to detail in every aspect of development.

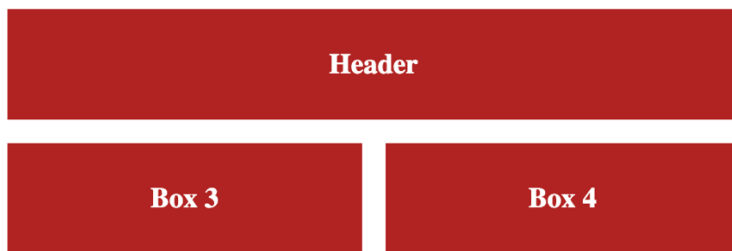
Color theory, typography, and layout design are *essential skills* for modern web developers.

22. Create the following form using HTML and CSS (10 points):



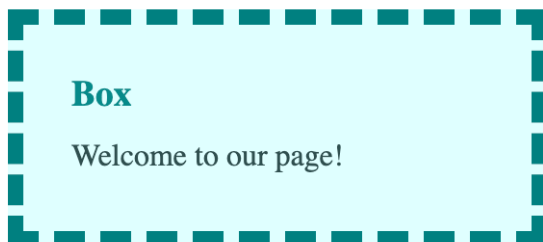
A login form with a yellow border. At the top, it says "Login Form" in bold. Below that are three input fields: "Username", "Password", and "Select Department" (a dropdown menu). At the bottom is a yellow "Sign In" button.

23. Create the following layout using HTML and CSS grid (10 points):



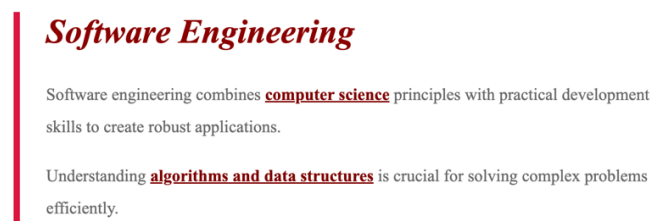
A layout with a red header bar labeled "Header". Below it are two red boxes, "Box 3" and "Box 4", side-by-side.

24. Use the box-model properties to create the following web page fragment (10 points):



A light blue box with a dashed teal border. Inside, it says "Box" in bold and "Welcome to our page!" below it.

25. Create the page below using HTML and CSS (10 points):



A page with a red vertical bar on the left. The title "Software Engineering" is in bold, italicized red. Below it, two paragraphs of text are shown: "Software engineering combines computer science principles with practical development skills to create robust applications." and "Understanding algorithms and data structures is crucial for solving complex problems efficiently."

26. Create the following form using HTML and CSS (10 points):

Email:

Age:

Comments:

Submit Form

27. Create the following layout using HTML and CSS grid (10 points):

Task 1

Task 2

Task 3

Task 4

28. Use the box-model properties to create the following web page fragment (10 points):

Border Design

Lorem ipsum dolor sit amet
consectetur adipisicing elit.
Consequuntur similique aliquid, unde
magnam at sit illum tempore

29. Create the page below using HTML and CSS (10 points):

Digital Innovation

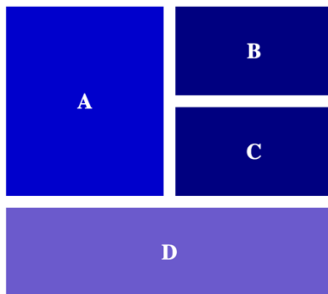
The digital age has brought revolutionary changes in how we communicate and share information across the globe.

Technologies like artificial intelligence and machine learning are transforming industries and creating new opportunities.

30. Create the following form using HTML and CSS (10 points):

Registration

31. Create the following layout using HTML and CSS grid (10 points):



32. On your website, the font size for the paragraph element should normally be 16px. Write media queries that change the font size to 18px when the screen width is greater than 480px, and to 20px when it's greater than 1024px. (10 points)

33. Create the page below using HTML and CSS (10 points):

Global Education

Education is the foundation of **personal growth** and societal development across all cultures and nations.

Access to quality learning resources and **skilled educators** remains essential for student success.

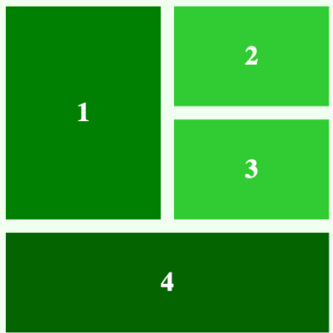
34. Create the following form using HTML and CSS (10 points):

Contact Us
Full Name:

Subject:

Message:

35. Create the following layout using HTML and CSS grid (10 points):



36. Use the **box-model** properties to create the following web page fragment: (10 points)



37. Create the page below using HTML and CSS (10 points):

Environmental Awareness

Protecting our planet requires **collective action** and commitment from individuals, communities, and governments worldwide.

Sustainable practices and **renewable energy** are essential for future generations.

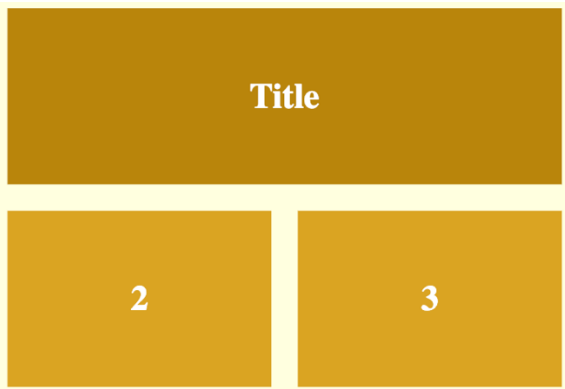
38. Create the following form using HTML and CSS (10 points):

Feedback Form
Name:

Phone:

Company:

39. Create the following layout using HTML and CSS grid (10 points):



40. Use the **box-model** properties to create the following web page fragment: (10 points)



41. Explain the key differences between HTTP/1.1, HTTP/2, and HTTP/3 in terms of connection handling. Why does each version improve upon the previous one? **(10 points)**
42. Describe the complete DNS resolution process from browser to authoritative DNS servers **(10 points)**
43. Explain the complete process of how a message travels from one computer to another using IP and TCP **(10 points)**
44. Describe the three main security features of HTTPS: Encryption, Data Integrity, and Authentication. Explain the complete HTTPS handshake process including SSL/TLS **(10 points)**
45. What are the three main security features of HTTPS?